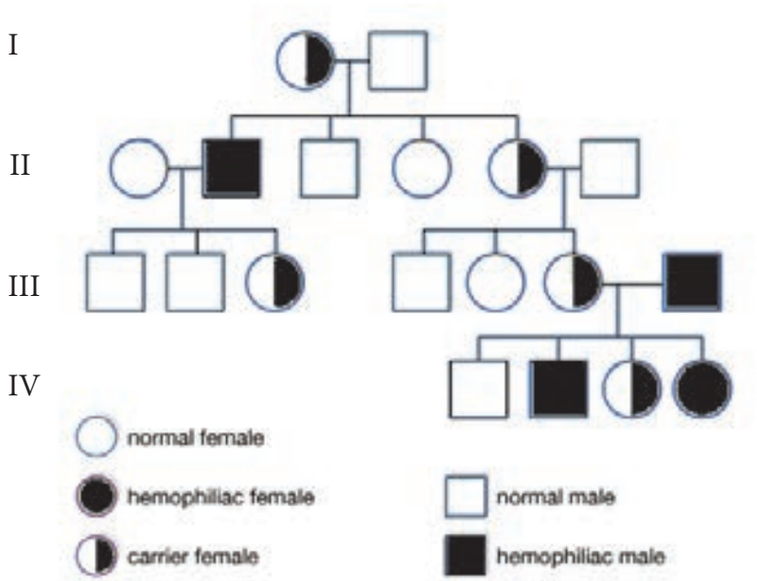
20. X – LINKED DISEASE - HAEMOPHILIA (BLEEDER'S DISEASE)

IDENTIFICATION

The given pedigree chart is identified as the genetic disease Haemophilia.

COMMENTS

1. Haemophilia or bleeder's disease (Royal disease) is the most notorious of all sex-linked diseases. The person suffering from this disease bleeds for a long period (30 minutes to 24 hours) during injury due to the failure of blood coagulation.
2. It is caused by a recessive X – linked gene more common in men than women.
3. The females are carriers of the disease and would transmit the disease to 50% of their sons even if the male parent is normal.
4. It follows criss – cross or zig – zag pattern of inheritance (i.e., grandfather transmits his X linked character to his grandson through carrier daughter).



DISCUSSION QUESTIONS

1. Observe the given pedigree chart and identify the affected individuals and carriers in the II generation.
2. Why are men affected often in X linked inheritance?
3. What is the pattern of inheritance in the given pedigree chart?
4. Why are women said to be carriers in X linked inheritance?
5. How is haemophilia caused?

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